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NON ORIENTED ELECTRICAL STEEL SHEET, IRON CORE, MANUFACTURING METHOD OF IRON CORE, MOTOR, AND MANUFACTURING METHOD OF MOTOR

Abstract

A non oriented electrical steel sheet includes, as a chemical composition, by mass %, 1.0% or more and 5.0% or less of Si, wherein a sheet thickness is 0.10 mm or more and 0.35 mm or less, an average grain size is 30 μm or more and 200 μm or less, an X value defined by $X = (2 \times B.\text{sub}.50L + B.\text{sub}.50C) / (3 \times I.\text{sub}.S)$ is 0.800 or more, and an iron loss $W.\text{sub}.10/1k$ is 80 W/kg or less.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of copending application Ser. No. 18/464,879, filed on Sep. 11, 2023, which is Continuation of application Ser. No. 18/077,789, filed on Dec. 8, 2022 (now U.S. Pat. No. 11,814,710, issued on Nov. 14, 2023), which is a Continuation of PCT International Application No. PCT/JP2022/029067, filed on Jul. 28, 2022, which claims priority under 35 U.S.C. 119 (a) to Patent Application No. 2021-126289, filed in Japan on Jul. 30, 2021, all of which are hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

[0002] The present invention relates to a non oriented electrical steel sheet. More specifically, the present invention relates to a non oriented electrical steel sheet, which is suitable for integrally punched iron cores of motors for electric vehicles, hybrid vehicles, or the like, an iron core, a manufacturing method of the iron core, a motor, and a manufacturing method of the motor.

[0003] Priority is claimed on Japanese Patent Application No. 2021-126289 filed on Jul. 30, 2021, the content of which is incorporated herein by reference.

BACKGROUND ART

[0004] Due to a need to reduce global warming gases, products with less energy consumption have been developed in industrial fields. For instance, in a field of an automobile, there are fuel-efficient vehicles such as hybrid-driven vehicles that combine a gasoline engine and a motor, and motor-

driven electric vehicles. A technology common to these fuel-efficient vehicles is a motor, and increasing an efficiency of the motor has become an important technology.

[0005] In General, a motor includes a stator and a rotor. The stator includes an iron core, and the iron core is classified as an integrally punched iron core and a segmented iron core. For the integrally punched iron core and the segmented iron core, there is a demand for a non oriented electrical steel sheet having excellent magnetic characteristics in a rolling direction (hereinafter referred to as “L direction”) and in a transverse direction (hereinafter referred to as “C direction”).

[0006] In addition, the motor shows excellent performance in a case where a gap between the stator and the rotor becomes smaller as an internal structure of the motor. Thus, each component of the motor is required to have a high shape accuracy. For instance, the integrally punched iron core and the segmented iron core are both formed by punching a steel sheet blank. However, in the integrally punched iron core, since the steel sheet blank is punched to be a hollow disc shape, the shape accuracy after punching may be deteriorated due to mechanical anisotropy of the steel sheet blank. Therefore, for the integrally punched iron core, it is desired for the non oriented electrical steel sheet to have small mechanical anisotropy.

[0007] For instance, Patent Document 1 discloses a technique related to a non oriented electrical steel sheet having excellent magnetic characteristics. Patent Document 2 discloses a technique related to a non oriented electrical steel sheet that can improve efficiency of a motor including a segmented iron core. Patent Document 3 discloses a technique related to a non oriented electrical steel sheet having excellent magnetic characteristics.

RELATED ART DOCUMENTS

Patent Documents

[0008] [Patent Document 1] Japanese Patent (Granted) Publication No. 5447167 [0009] [Patent Document 2] Japanese Patent (Granted) Publication No. 5716315 [0010] [Patent Document 3] PCT International Publication No. WO2013/069754

SUMMARY OF INVENTION

Technical Problem to be Solved

[0011] The present invention has been made in consideration of the above mentioned situations. An object of the invention is to provide a non oriented electrical steel sheet with excellent magnetic characteristics and small mechanical anisotropy for an integrally punched iron core, an iron core, a manufacturing method of the iron core, a motor, and a manufacturing method of the motor.

Solution to Problem

[0012] An aspect of the present invention employs the following.

[0013] (1) A non oriented electrical steel sheet according to an aspect of the present invention includes a chemical composition containing, by mass %, [0014] 0.005% or less of C, [0015] 1.0% or more and 5.0% or less of Si, [0016] less than 2.5% of sol. Al, [0017] 3.0% or less of Mn, [0018] 0.3% or less of P, [0019] 0.01% or less of S, [0020] 0.01% or less of N, [0021] 0.10% or less of B, [0022] 0.10% or less of O, [0023] 0.10% or less of Mg, [0024] 0.01% or less of Ca, [0025] 0.10% or less of Ti, [0026] 0.10% or less of V, [0027] 5.0% or less of Cr, [0028] 5.0% or less of Ni, [0029] 5.0% or less of Cu, [0030] 0.10% or less of Zr, [0031] 0.10% or less of Sn, [0032] 0.10% or less of Sb, [0033] 0.10% or less of Ce, [0034] 0.10% or less of Nd, [0035] 0.10% or less of Bi, [0036] 0.10% or less of W, [0037] 0.10% or less of Mo, [0038] 0.10% or less of Nb, [0039] 0.10% or less of Y, and [0040] a balance consisting of Fe and impurities, wherein [0041] a sheet thickness is 0.10 mm or more and 0.35 mm or less, [0042] an average grain size is 30 μm or more and 200 μm or less, [0043] an X value defined by the following expression 1 is 0.800 or more, and [0044] an iron loss $W_{\text{sub.10/1k}}$ when excited so as to be a magnetic flux density of 1.0 T at a frequency of 1 kHz is 80 W/kg or less, [0045] where the expression 1 is $X = (2 \times B_{\text{sub.50L}} + B_{\text{sub.50C}}) / (3 \times I_{\text{sub.S}})$ and [0046] where $B_{\text{sub.50L}}$ denotes a magnetic flux density in a rolling direction when magnetized with a magnetizing force of 5000 A/m, $B_{\text{sub.50C}}$ denotes a magnetic flux density in a transverse direction when magnetized with a magnetizing force of 5000 A/m, and $I_{\text{sub.S}}$ denotes a spontaneous magnetization at room temperature.

[0047] (2) In the non oriented electrical steel according to (1), the chemical composition may include, by mass %, more than 3.25% and 5.0% or less of Si.

[0048] (3) In the non oriented electrical steel according to (1) or (2), the chemical composition may include, by mass %, at least one of [0049] 0.0010% or more and 0.005% or less of C, [0050] 0.10% or more and less than 2.5% of sol. Al, [0051] 0.0010% or more and 3.0% or less of Mn, [0052] 0.0010% or more and 0.3% or less of P, [0053] 0.0001% or more and 0.01% or less of S, [0054] 0.0015% or more and 0.01% or less of N, [0055] 0.0001% or more and 0.10% or less of B, [0056] 0.0001% or more and 0.10% or less of O, [0057] 0.0001% or more and 0.10% or less of Mg, [0058] 0.0003% or more and 0.01% or less of Ca, [0059] 0.0001% or more and 0.10% or less of Ti, [0060] 0.0001% or more and 0.10% or less of V, [0061] 0.0010% or more and 5.0% or less of Cr, [0062] 0.0010% or more and 5.0% or less of Ni, [0063] 0.0010% or more and 5.0% or less of Cu, [0064] 0.0002% or more and 0.10% or less of Zr, [0065] 0.0010% or more and 0.10% or less of Sn, [0066] 0.0010% or more and 0.10% or less of Sb, [0067] 0.001% or more and 0.10% or less of Ce, [0068] 0.002% or more and 0.10% or less of Nd, [0069] 0.002% or more and 0.10% or less of Bi, [0070] 0.002% or more and 0.10% or less of W, [0071] 0.002% or more and 0.10% or less of Mo, [0072] 0.0001% or more and 0.10% or less of Nb, and [0073] 0.002% or more and 0.10% or less of Y.

[0074] (4) In the non oriented electrical steel according to any one of (1) to (3), the chemical composition may include, by mass %, more than 4.0% in total of Si and sol. Al.

[0075] (5) In the non oriented electrical steel according to any one of (1) to (4), the X value may be 0.800 or more and less than 0.845.

[0076] (6) In the non oriented electrical steel according to any one of (1) to (5), the X value may be 0.800 or more and less than 0.830.

[0077] (7) An iron core according to an aspect of the present invention may include the non oriented electrical steel sheet according to any one of (1) to (6).

[0078] (8) A manufacturing method of an iron core according to an aspect of the present invention may include a process of punching and laminating the non oriented electrical steel sheet according to any one of (1) to (6).

[0079] (9) A motor according to an aspect of the present invention may include the iron core according to (7).

[0080] (10) A manufacturing method of a motor may include [0081] a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to any one of (1) to (6) and [0082] a process of assembling the motor using the iron core.

Effects of Invention

[0083] According to the above aspects of the present invention, it is possible to provide the non oriented electrical steel sheet with excellent magnetic characteristics and small mechanical anisotropy for the integrally punched iron core, the iron core, the manufacturing method of the iron core, the motor, and the manufacturing method of the motor.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0084] FIG. 1 is a schematic illustration of a non oriented electrical steel sheet according to an embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0085] Hereinafter, a preferable embodiment of the present invention is described in detail. However, the present invention is not limited only to the configuration which is disclosed in the embodiment, and various modifications are possible without departing from the aspect of the present invention. In addition, the limitation range as described below includes a lower limit and an upper limit thereof. However, the value expressed by “more than” or “less than” does not include in the limitation range. “%” of the amount of respective elements expresses “mass %”.

[0086] FIG. 1 shows a schematic illustration of a non oriented electrical steel sheet according to the embodiment of the present invention. (Chemical Composition)

[0087] Limitation reasons of the chemical composition of the non oriented electrical steel sheet according to the present embodiment are explained.

[0088] As a chemical composition, the non oriented electrical steel sheet according to the present embodiment contains Si, optional elements as necessary, and a balance consisting of Fe and impurities. Hereinafter, each element is explained.

[0089] C: 0% or more and 0.005% or less

[0090] C (carbon) is an element contained as an impurity and deteriorates the magnetic characteristics. Thus, the C content is to be 0.005% or less. Preferably, the C content is 0.003% or less. Since it is preferable that the C content is low, a lower limit does not need to be limited, and the lower limit may be 0%. However, it is not easy to industrially control the content to be 0%, and thus, the lower limit may be more than 0% or 0.0010%.

[0091] Si: 1.0% or more and 5.0% or less

[0092] Si (silicon) is an element that is effective in increasing electrical resistivity of the steel sheet and reducing iron loss. Thus, the Si content is to be 1.0% or more. Moreover, Si is an effective element for the non oriented electrical steel sheet for the integrally punched iron core to achieve both magnetic characteristics and mechanical anisotropy. In this case, the Si content is preferably more than 3.25%, more preferably 3.27% or more, further more preferably 3.30% or more, and further more preferably 3.40% or more. On the other hand, when the Si content is excessive, a magnetic flux density deteriorates significantly. Thus, the Si content is to be 5.0% or less. The Si content is preferably 4.0% or less, and more preferably 3.5% or less.

[0093] sol. Al: 0% or more and less than 2.5%

[0094] Al (aluminum) is an optional element that is effective in increasing the electrical resistivity of the steel sheet and reducing the iron loss. However, when the content is excessive, the magnetic flux density deteriorates significantly. Thus, the sol. Al content is to be less than 2.5%. A lower limit of sol. Al does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the sol. Al content is preferably 0.10% or more. Herein, the sol. Al expresses acid-soluble aluminum.

[0095] Moreover, Si and Al are elements effective in achieving both magnetic characteristics and mechanical anisotropy. Thus, the total amount of Si and sol. Al is preferably more than 4.0%, more preferably more than 4.10%, and further more preferably more than 4.15%. On the other hand, Si and Al have a strong effect on solid solution strengthening. When the content is excessive, cold rolling becomes difficult to be performed. Thus, the total amount of Si and sol. Al is preferably less than 5.5%.

[0096] Mn: 0% or more and 3.0% or less

[0097] Mn (manganese) is an optional element that is effective in increasing the electrical resistivity of the steel sheet and reducing the iron loss. However, since an alloying cost of is higher than Si or Al, an increase in the Mn content is economically disadvantageous. Thus, the Mn content is to be 3.0% or less. Preferably, the Mn content is 2.5% or less. A lower limit of Mn does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Mn content is preferably 0.0010% or more, and more preferably 0.010% or more.

[0098] P: 0% or more and 0.3% or less

[0099] P (phosphorus) is an element generally contained as an impurity. However, P has an effect in improving texture of the non oriented electrical steel sheet and thereby improving the magnetic characteristics. Thus, P may be included as necessary. However, P is a solid solution strengthening element. When the P content is excessive, the steel sheet is hardened and thereby the cold rolling becomes difficult to be performed. Thus, the P content is to be 0.3% or less. The P content is preferably 0.2% or less. A lower limit of P does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the P content is preferably 0.0010% or more, and more preferably 0.015% or more.

[0100] S: 0% or more and 0.01% or less

[0101] S (sulfur) is contained as an impurity and forms fine MnS by bonding to Mn in steel. As a result, S suppresses grain growth during annealing and deteriorates the magnetic characteristics of the non oriented electrical steel sheet. Thus, the S content is to be 0.01% or less. The S content is preferably 0.005% or less, and more preferably 0.003% or less. Since it is preferable that the S content is low, a lower limit does not need to be limited, and the lower limit may be 0%. However, it is not easy to industrially control the content to be 0%, and thus, the lower limit may be 0.0001%.

[0102] N: 0% or more and 0.01% or less

[0103] N (nitrogen) is contained as an impurity and forms fine AlN by bonding to Al in steel. As a result, N suppresses the grain growth during annealing and deteriorates the magnetic characteristics. Thus, the N content is to be 0.01% or less. The N content is preferably 0.005% or less, and more preferably 0.003% or less. Since it is preferable that the N content is low, a lower limit does not need to be limited, and the lower limit may be 0%. However, it is not easy to industrially control the content to be 0%, and thus, the lower limit may be 0.0001% or more, may be more than 0.0015%, or may be 0.0025% or more.

[0104] Sn: 0% or more and 0.10% or less

[0105] Sb: 0% or more and 0.10% or less

[0106] Sn (tin) and Sb (antimony) are optional elements having effect in improving the texture of the non oriented electrical steel sheet and thereby improving the magnetic characteristics (for instance, magnetic flux density). Thus, Sn and Sb may be included as necessary. However, when the content is excessive, the steel may become brittle and fracture may occur during cold rolling. Moreover, the magnetic characteristics are deteriorated. Thus, the Sn content and the Sb content are to be 0.10% or less, respectively. Lower limits of Sn and Sb do not need to be limited, and the lower limits may be 0%. In order to reliably obtain the above effect, the Sn content is preferably 0.0010% or more, and more preferably 0.01% or more. Moreover, the Sb content is preferably 0.0010% or more, more preferably 0.002% or more, further more preferably 0.01% or more, and further more preferably more than 0.025%.

[0107] Ca: 0% or more and 0.01% or less

[0108] Ca (calcium) is an optional element that suppresses precipitation of fine sulfides (MnS, Cu.sub.2S, or the like) by forming coarse sulfides. When the Ca content is favorable, inclusions are controlled, the grain growth is improved during annealing, and thereby, the magnetic characteristics (for instance, iron loss) are improved. However, when the content is excessive, the effect thereof is saturated, and the cost increases. Thus, the Ca content is to be 0.01% or less. The Ca content is preferably 0.008% or less, and more preferably 0.005% or less. A lower limit of Ca does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Ca content is preferably 0.0003% or more. The Ca content is preferably 0.001% or more, and more preferably 0.003% or more.

[0109] Cr: 0% or more and 5.0% or less

[0110] Cr (chromium) is an optional element that increases the electrical resistivity and improves the magnetic characteristics (for instance, iron loss). However, when the content is excessive, a saturation magnetic flux density may decrease, the effect thereof is saturated, and the cost increases. Thus, the Cr content is to be 5.0% or less. The Cr content is preferably 0.5% or less, and more preferably 0.1% or less. A lower limit of Cr does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Cr content is preferably 0.0010% or more.

[0111] Ni: 0% or more and 5.0% or less

[0112] Ni (nickel) is an optional element that improves the magnetic characteristics (for instance, saturation magnetic flux density). However, when the content is excessive, the effect thereof is saturated, and the cost increases. Thus, the Ni content is to be 5.0% or less. The Ni content is preferably 0.5% or less, and more preferably 0.1% or less. A lower limit of Ni does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Ni content is preferably 0.0010% or more.

[0113] Cu: 0% or more and 5.0% or less

[0114] Cu (copper) is an optional element that improves strength of the steel sheet. However, when the content is excessive, the saturation magnetic flux density may decrease, the effect thereof is saturated, and the cost increases. Thus, the Cu content is to be 5.0% or less. The Cu content is preferably 0.1% or less. A lower limit of Cu does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Cu content is preferably 0.0010% or more.

[0115] Ce: 0% or more and 0.10% or less

[0116] Ce (cerium) is an optional element that suppresses the precipitation of fine sulfides (MnS, Cu.sub.2S, or the like) by forming coarse sulfides coarse oxysulfides, or the like. As a result, the grain growth is improved, and the iron loss is improved. However, when the content is excessive, the iron loss may be deteriorated by forming oxides in addition to sulfides and oxysulfides, and the effect thereof is saturated, and the cost increases. Thus, the Ce content is to be 0.10% or less. The Ce content is preferably 0.01% or less, more preferably 0.009% or less, and further more preferably 0.008% or less. A lower limit of Ce does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Ce content is preferably 0.001% or more. The Ce content is more preferably 0.002% or more, more preferably 0.003% or more, and further more preferably 0.005% or more.

[0117] In addition to the above elements, the non oriented electrical steel sheet according to the present embodiment may contain, as a chemical composition, the optional elements such as B, O, Mg, Ti, V, Zr, Nd, Bi, W, Mo, Nb, and Y. Amounts of these optional elements may be controlled on the basis of known knowledge. For instance, the amounts of these optional elements may be as follows. [0118] B: 0% or more and 0.10% or less [0119] O: 0% or more and 0.10% or less [0120] Mg: 0% or more and 0.10% or less [0121] Ti: 0% or more and 0.10% or less [0122] V: 0% or more and 0.10% or less [0123] Zr: 0% or more and 0.10% or less [0124] Nd: 0% or more and 0.10% or less [0125] Bi: 0% or more and 0.10% or less [0126] W: 0% or more and 0.10% or less [0127] Mo: 0% or more and 0.10% or less [0128] Nb: 0% or more and 0.10% or less [0129] Y: 0% or more and 0.10% or less

[0130] Moreover, the non oriented electrical steel sheet according to the present embodiment may contain, as a chemical composition, by mass %, at least one of [0131] 0.0010% or more and 0.005% or less of C, [0132] 0.10% or more and less than 2.5% of sol. Al, [0133] 0.0010% or more and 3.0% or less of Mn, [0134] 0.0010% or more and 0.3% or less of P, [0135] 0.0001% or more and 0.01% or less of S, [0136] 0.0015% or more and 0.01% or less of N, [0137] 0.0001% or more and 0.10% or less of B, [0138] 0.0001% or more and 0.10% or less of O, [0139] 0.0001% or more and 0.10% or less of Mg, [0140] 0.0003% or more and 0.01% or less of Ca, [0141] 0.0001% or more and 0.10% or less of Ti, [0142] 0.0001% or more and 0.10% or less of V, [0143] 0.0010% or more and 5.0% or less of Cr, [0144] 0.0010% or more and 5.0% or less of Ni, [0145] 0.0010% or more and 5.0% or less of Cu, [0146] 0.0002% or more and 0.10% or less of Zr, [0147] 0.0010% or more and 0.10% or less of Sn, [0148] 0.0010% or more and 0.10% or less of Sb, [0149] 0.001% or more and 0.10% or less of Ce, [0150] 0.002% or more and 0.10% or less of Nd, [0151] 0.002% or more and 0.10% or less of Bi, [0152] 0.002% or more and 0.10% or less of W, [0153] 0.002% or more and 0.10% or less of Mo, [0154] 0.0001% or more and 0.10% or less of Nb, and [0155] 0.002% or more and 0.10% or less of Y.

[0156] The B content is preferably 0.01% or less, the O content is preferably 0.01% or less, the Mg content is preferably 0.005% or less, the Ti content is preferably 0.002% or less, the V content is preferably 0.002% or less, the Zr content is preferably 0.002% or less, the Nd content is preferably 0.01% or less, the Bi content is preferably 0.01% or less, the W content is preferably 0.01% or less, the Mo content is preferably 0.01% or less, the Nb content is preferably 0.002% or less, and the Y content is preferably 0.01% or less. Moreover, the Ti content is preferably 0.001% or more, the V content is preferably 0.002% or more, and the Nb content is preferably 0.002% or more.

[0157] The chemical composition as described above may be measured by typical analytical methods for the steel. For instance, the chemical composition may be measured by using ICP-AES (Inductively Coupled Plasma-Atomic Emission Spectrometer: inductively coupled plasma emission spectroscopy spectrometry). Herein, the acid soluble Al may be measured by ICP-AES using filtrate after heating and dissolving the sample in acid. In addition, C and S may be measured by the infrared absorption method after combustion, N may be measured by the thermal conductometric method after fusion in a current of inert gas, and O may be measured by, for instance, the non-dispersive infrared absorption method after fusion in a current of inert gas.

[0158] The above chemical composition is that of the non oriented electrical steel sheet without insulation coating. When the non oriented electrical steel sheet to be the measurement sample has the insulation coating and the like on the surface, the chemical composition is measured after removing the coating. For instance, the insulation coating may be removed by the following method. First, the non oriented electrical steel sheet having the insulation coating and the like is immersed in sodium hydroxide aqueous solution, sulfuric acid aqueous solution, and nitric acid aqueous solution in this order. The steel sheet after the immersion is washed. Finally, the steel sheet is dried with warm air. Thereby, it is possible to obtain the non oriented electrical steel sheet from which the insulation coating is removed. Alternatively, the insulation coating may be removed by grinding.

(Magnetic Characteristics)

[0159] As the magnetic flux density, an X value defined by the following expression 1 is to be 0.800 or more. In order to improve the magnetic characteristics, the X value is preferably 0.820 or more.

[00001] $X = (2 \times B_{50L} + B_{50C}) / (3 \times I_S)$ Expression1

[0160] Herein, [0161] B.sub.50L denotes the magnetic flux density in a rolling direction when magnetized with magnetizing force of 5000 A/m, [0162] B.sub.50C denotes the magnetic flux density in a transverse direction when magnetized with magnetizing force of 5000 A/m, and [0163] I.sub.S denotes spontaneous magnetization at room temperature.

[0164] I.sub.S in the expression 1 may be obtained by the following expression 2 and expression 3. The expression 2 is for obtaining the spontaneous magnetization assuming that the spontaneous magnetization of the steel sheet is simply attenuated by elements other than Fe. Density of the steel sheet in the expression 2 may be measured on the basis of JIS Z 8807:2012. In a case where the insulation coating is applied, the density may be measured by the above described method under condition such that the insulation coating exists, and the same value of the density is also used at the time of evaluating the magnetic characteristics described later. Density of Fe in the expression 2 may be 7.873 g/cm.sup.3.

[00002] $I_S = 2.16 \times \{(\text{densityofsteelsheet}) / (\text{densityofFe})\} \times [\text{Fecontent}(\text{mass}\%)] / 100$ Expression2

Fecontent(mass%) = 100(mass%) - [totalamount (mass%) of C, Si, Mn, sol. Al, P, S, N, B, O, Mg, Ca, Ti, V, Cr, Ni, Cu, Zr, Sn, Sb, Ce, Nd, Bi, W, Mo, Nb, and Y]

[0165] In order to achieve both magnetic characteristics and mechanical anisotropy as the non oriented electrical steel sheet for the integrally punched iron core, the X value is preferably less than 0.845, more preferably less than 0.840, further more preferably less than 0.835, and further more preferably less than 0.830.

[0166] As the iron loss, an iron loss W.sub.10/1k when excited so as to have the magnetic flux density of 1.0 T at frequency of 1 kHz is to be 80 W/kg or less. The iron loss W.sub.10/1k is preferably 70 W/kg or less, and more preferably 49 W/kg or less. Although a lower limit of the iron loss W.sub.10/1k does not need to be limited, the lower limit may be 30 W/kg as necessary.

[0167] The magnetic characteristics may be measured on the basis of the single sheet tester (SST) method regulated by JIS C 2556:2015. Instead of taking a test piece with size regulated by JIS, a test piece with smaller size, for instance, a test piece of width 55 mm×length 55 mm, may be taken and measured on the basis of the single sheet tester. In a case where the test piece of width 55 mm×length 55 mm is hardly taken, the measurement based on the single sheet tester may be performed using two test pieces of width 8 mm×length 16 mm as a test piece of width 16 mm×length 16 mm. At that time, it is preferable to use an Epstein equivalent value which is converted so as to correspond to a measurement value with an Epstein tester regulated in JIS C 2550:2011.

(Average Grain Size)

[0168] When grain size is excessively coarse or fine, the iron loss under high frequency may deteriorates. Thus, the average grain size is to be 30 μm

or more and 200 μm or less.

[0169] The average grain size may be measured on the basis of an intercept method regulated by JIS G 0551:2020. For instance, in a longitudinal sectional micrograph, an average value of grain sizes may be measured by the intercept method along sheet thickness direction and rolling direction. As the longitudinal sectional micrograph, an optical micrograph may be used, and for instance, a micrograph obtained at a magnification of 50-fold may be used.

(Sheet Thickness)

[0170] Sheet thickness is to be 0.35 mm or less. The sheet thickness is preferably 0.30 mm or less. On the other hand, when the sheet thickness is excessively thin, productivity of the steel sheet and motor deteriorates significantly. Thus, the sheet thickness is to be 0.10 mm or more. The sheet thickness is preferably 0.15 mm or more.

[0171] The sheet thickness may be measured by a micrometer. When the non oriented electrical steel sheet to be the measurement sample has the insulation coating and the like on the surface, the sheet thickness is measured after removing the coating. The method for removing the insulation coating is as described above.

[0172] In addition, as mechanical anisotropy, a roundness after punching to be true circle is preferably more than 0.9997 and 1.0000 or less. Specifically, when the punching is conducted using a die with a hollow disc shape of an inner diameter of 80.0 mm and an outer diameter of 100 mm and when 60 sheets of punched pieces are laminated and fastened to be formed, a value (roundness) obtained by dividing a minimum value of a diameter of an inner circumference of the formed piece by a maximum value of a diameter of the inner circumference of the formed piece is preferably more than 0.9997 and 1.0000 or less.

[0173] When the above roundness is more than 0.9997, a shape accuracy of the punched piece can be regarded as high. As a result, when it is used for the motor, an increase in cogging torque and an increase in vibration noise can be favorably suppressed. The above roundness is preferably more than 0.9998, and more preferably 0.9999 or more.

[0174] The roundness may be measured by the following method. The non oriented electrical steel sheet is punched using the die with the hollow disc shape (true circle) of the inner diameter of 80.0 mm and the outer diameter of 100 mm at a punching speed of 250 strokes/min by a 25 t continuous progressive press-working apparatus. The 60 sheets of punched pieces are laminated and fastened to form the core. The obtained ring-shaped core simulates the integrally punched iron core for the stator of the motor, and the roundness of the inner circumference can be used as an index of accuracy of an air gap with a rotor core. Diameters of the inner circumference of the obtained ring-shaped core are measured at plural positions, and a ratio of a minimum value to a maximum value of the measured diameters is regarded as the roundness. Specifically, when the punching is conducted using the die with the hollow disc shape (true circle) of the inner diameter of 80.0 mm and the outer diameter of 100 mm and when the 60 sheets of punched pieces are laminated and fastened to be formed, the value obtained by dividing the minimum value of the diameter of the inner circumference of the formed piece by the maximum value of the diameter of the inner circumference of the formed piece is regarded as the roundness.

[0175] The non oriented electrical steel sheet according to the present embodiment is excellent in both the magnetic characteristics and the roundness for the integrally punched iron core. For instance, the non oriented electrical steel sheet according to the present embodiment satisfies the X value of 0.800 or more and the iron loss $W_{\text{sub.10/1k}}$ of 80 W/kg or less, and as a result, it is possible to obtain the effect such that the roundness is excellent. Moreover, when the chemical composition and manufacturing conditions are favorably controlled, the non oriented electrical steel sheet according to the present embodiment satisfies the X value of 0.800 or more and less than 0.845 and the iron loss $W_{\text{sub.10/1k}}$ of 49 W/kg or less, and as a result, it is possible to obtain the effect such that the roundness is more excellent. In this case, it can be judged that the magnetic characteristics and the roundness are simultaneously and preferably achieved for the integrally punched iron core.

(Iron Core and Motor)

[0176] Since the non oriented electrical steel sheet according to the present embodiment has excellent magnetic characteristics and small mechanical anisotropy, it is suitable for the integrally punched iron core of motor for electric vehicles, hybrid vehicles, or the like. Thus, an iron core including the non oriented electrical steel sheet according to the present embodiment exhibits excellent performance. Moreover, since the non oriented electrical steel sheet according to the present embodiment is suitable for the integrally punched iron core, a motor including the iron core exhibits excellent performance.

(Manufacturing Method)

[0177] Hereinafter, an instance of a manufacturing method of the non oriented electrical steel sheet according to the present embodiment is explained below. The non oriented electrical steel sheet according to the present embodiment is not particularly limited in the manufacturing method as long as the above features are included. The following manufacturing method is an instance for manufacturing the non oriented electrical steel sheet according to the present embodiment and a favorable manufacturing method of the non oriented electrical steel sheet according to the present embodiment.

[0178] The manufacturing method of the non oriented electrical steel sheet according to the present embodiment includes the following processes (A) to (D). [0179] (A) A first cold rolling process of subjecting a hot rolled steel sheet having the chemical composition described above to cold rolling under conditions such that a rolling reduction is 10% or larger and 75% or smaller. [0180] (B) An intermediate annealing process of subjecting the cold rolled steel sheet obtained in the first cold rolling process to intermediate annealing under condition such that an average heating rate from 500° C. to 650° C. is 30° C./hour or faster and 1000° C./second or slower (0.0083° C./second or faster and 1000° C./second or slower), a retention temperature is 700° C. or higher and 1100° C. or lower, and a retention time is 10 seconds or longer and 40 hours or shorter (0.0028 hours or longer and 40 hours or shorter). [0181] (C) A second cold rolling process of subjecting the intermediate-annealed steel sheet obtained in the intermediate annealing process to cold rolling under conditions such that a rolling reduction is 50% or larger and 85% or smaller to obtain a sheet thickness of 0.10 mm or more and 0.35 mm or less. [0182] (D) A final annealing process of subjecting the cold rolled steel sheet obtained in the second cold rolling process to final annealing under conditions such that a temperature range to be retained is 900° C. or higher and 1200° C. or lower.

[0183] Each process is explained below.

(First Cold Rolling Process)

[0184] In the first cold rolling process, the hot rolled steel sheet having the above chemical composition is subjected to cold rolling at the rolling reduction (cumulative rolling reduction) of 10% or larger and 75% or smaller.

[0185] When the rolling reduction in the first cold rolling process is smaller than 10% or larger than 75%, the intended magnetic characteristics and roundness may not be obtained. Thus, the rolling reduction in the first cold rolling process is to be 10% or larger and 75% or smaller.

[0186] Conditions of the cold rolling other than the above, such as a steel sheet temperature during cold rolling and a diameter of the rolling roll, are not particularly limited, and are appropriately selected depending on the chemical composition of the hot rolled steel sheet, the intended sheet thickness of the steel sheet, or the like.

[0187] In general, the hot rolled steel sheet is subject to cold rolling after a scale formed on a surface of the steel sheet during hot rolling is removed by pickling. As described later, in a case where the hot rolled steel sheet is subjected to hot-band annealing, the hot rolled steel sheet may be pickled either before the hot-band annealing or after the hot-band annealing.

(Intermediate Annealing Process)

[0188] In the intermediate annealing process, the cold rolled steel sheet obtained in the above first cold rolling process is subjected to intermediate annealing under condition such that an average heating rate from 500° C. to 650° C. is 30° C./hour or faster and 1000° C./second or slower (0.0083°

C./second or faster and 1000° C./second or slower), a retention temperature is 700° C. or higher and 1100° C. or lower, and a retention time is 10 seconds or longer and 40 hours or shorter (0.0028 hours or longer and 40 hours or shorter).

[0189] When the above conditions are not satisfied in the intermediate annealing process, the intended magnetic characteristics and roundness may not be obtained. Conditions of the intermediate annealing other than the above are not particularly limited.

[0190] The average heating rate from 500° C. to 650° C. is preferably 300° C./second or faster. The retention temperature is preferably 850° C. or higher. The retention time is preferably 180 seconds or shorter (0.05 hours or shorter). In particular, when the Si content of more than 3.25%, the average heating rate from 500° C. to 650° C. of 300° C./second or faster, the retention temperature of 850° C. or higher, and the retention time of 180 seconds or shorter are simultaneously satisfied in addition to satisfying the conditions of the present embodiment, it is possible to obtain the non oriented electrical steel sheet in which both desirable magnetic characteristics and roundness are simultaneously and preferably achieved.

(Second Cold Rolling Process)

[0191] In the second cold rolling process, the intermediate-annealed steel sheet obtained in the above intermediate annealing process is subjected to cold rolling at the rolling reduction (cumulative rolling reduction) of 50% or larger and 85% or smaller to obtain the sheet thickness of 0.10 mm or more and 0.35 mm or less.

[0192] When the rolling reduction in the second cold rolling process is smaller than 50% or larger than 85%, the intended magnetic characteristics and roundness may not be obtained. Thus, the rolling reduction in the second cold rolling process is to be 50% or larger and 85% or smaller.

[0193] The sheet thickness is to be 0.10 mm or more and 0.35 mm or less. The sheet thickness is preferably 0.15 mm or more and 0.30 mm or less.

[0194] Conditions of the cold rolling other than the above, such as a steel sheet temperature during cold rolling and a diameter of the rolling roll, are not particularly limited, and are appropriately selected depending on the chemical composition of the steel sheet, the intended sheet thickness of the steel sheet, or the like.

(Final Annealing Process)

[0195] In the final annealing process, the cold rolled steel sheet obtained in the above second cold rolling process is subjected to final annealing at the temperature range to be retained of 900° C. or higher and 1200° C. or lower.

[0196] When the final annealing temperature in the final annealing process is lower than 900° C., the average grain size may become less than 30 μm due to insufficient grain growth, and thereby sufficient magnetic characteristics may not be obtained. Thus, the final annealing temperature is to be 900° C. or higher. On the other hand, when the final annealing temperature is higher than 1200° C., the grain growth may proceed excessively, the average grain size may become more than 200 μm, and thereby sufficient magnetic characteristics may not be obtained. Thus, the final annealing temperature is to be 1200° C. or lower.

[0197] The final annealing time for retaining the cold rolled steel sheet in the temperature range of 900° C. or higher and 1200° C. or lower may not be particularly specified, but it is preferably 1 second or longer to more reliably obtain favorable magnetic characteristics. On the other hand, from a productive standpoint, the final annealing time is preferably 120 seconds or shorter.

[0198] Conditions of the final annealing other than the above are not particularly limited.

(Hot-Band Annealing Process)

[0199] The hot rolled steel sheet to be subjected to the above first cold rolling process may be subjected to hot-band annealing. When the hot rolled steel sheet is subjected to hot-band annealing, it is possible to obtain favorable magnetic characteristics.

[0200] The hot-band annealing may be performed by either box annealing or continuous annealing. When the box annealing is performed, the hot rolled steel sheet is preferably retained in a temperature range of 700° C. or higher and 900° C. or lower for 1 hour or longer and 20 hours or shorter. When the continuous annealing is performed, the hot rolled steel sheet is preferably retained in a temperature range of 850° C. or higher and 1100° C. or lower for 1 second or longer and 180 seconds or shorter.

[0201] Conditions of the hot-band annealing other than the above are not particularly limited.

(Hot Rolling Process)

[0202] The hot rolled steel sheet to be subjected to the first cold rolling process can be obtained by subjecting a steel ingot or steel piece (hereinafter referred to as “slab”) having the above chemical composition to hot rolling.

[0203] In the hot rolling, a steel having the above chemical composition is made into the slab by typical methods such as continuous casting or blooming the steel ingot. The slab is put into a heating furnace and then subjected to hot rolling. At this time, when the slab temperature is high, the hot rolling may be performed without putting the slab into the heating furnace.

[0204] Conditions of the hot-band annealing are not particularly limited.

(Other Processes)

[0205] After the final annealing process, a coating process of applying an insulation coating including only an organic component, only an inorganic component, or an organic-inorganic compound to a surface of the steel sheet may be performed by typical methods. From a standpoint of reducing an environmental load, an insulation coating that does not include chromium may be applied. Moreover, the coating process may be a process of applying an insulation coating that is adhesiveness by heating and pressurizing. As coating material exhibiting adhesiveness, an acrylic resin, a phenol resin, an epoxy resin, a melamine resin, or the like can be used.

(Manufacturing Method of Iron Core and Manufacturing Method of Motor)

[0206] An integrally punched iron core may be manufactured using the non oriented electrical steel sheet according to the present embodiment obtained as described above. A manufacturing method of the iron core may include a process of punching and laminating the above non oriented electrical steel sheet. Moreover, a motor may be manufactured using the integrally punched iron core. A manufacturing method of the motor may include a process of preparing an iron core by punching and laminating the above non oriented electrical steel sheet, and a process of assembling the motor using the iron core.

Examples

[0207] The effects of an aspect of the present invention are described in detail with reference to the following examples. However, the condition in the examples is an example condition employed to confirm the operability and the effects of the present invention, so that the present invention is not limited to the example condition. The present invention can employ various types of conditions as long as the conditions do not depart from the scope of the present invention and can achieve the object of the present invention. Hereinafter, the present invention is explained in detail with reference to examples and comparative examples.

[0208] Non oriented electrical steel sheets were prepared by performing each process under the conditions shown in Tables 1 to 16 using slabs whose chemical compositions were adjusted. Moreover, in a case where hot-band annealing was not performed, pickling was performed after hot rolling. In a case where hot-band annealing was performed, pickling for Test No. 1 and 17 was performed before the hot-band annealing, and pickling for the others was performed after the hot-band annealing. Moreover, a retention time of the final annealing was 30 seconds.

[0209] A chemical composition, a sheet thickness, an average grain size, an X value related to a magnetic flux density, an iron loss W_{sub}.10/1k, and a roundness of the prepared non oriented electrical steel sheet were measured. Measurement methods thereof are as described above. Measurement results are shown in Tables 1 to 16. Herein, a chemical composition of the prepared non oriented electrical steel sheet was substantially the same as a chemical composition of the slab. The element represented by “-” in the table indicates that it was not consciously controlled and prepared.

Moreover, the Si content indicated by “3.3” in the table was more than 3.25%. Moreover, the manufacturing condition represented by “-” in the table indicates that it was not controlled. Moreover, a sheet thickness of the prepared non oriented electrical steel sheet was the same as a final sheet thickness after second cold rolling process.

0.002 4.8 7.503 1.949 37 S25 ———— 4.003 3.8 7.602 1.995 38 S26 0.002 ———— 3.9 7.596

1.991 39 S27 — 0.0005 ———— 4.0 7.590 1.988 40 S28 ———— 0.031 ———— 5.1 7.452 1.914 41 S29 ————

0.005 ———— 5.2 7.445 1.911 42 S30 ———— 0.003 ———— 5.0 7.477 1.925 43 S31 ———— 0.005 ————

— 4.0 7.569 1.974 44 S32 ———— 0.003 ———— 4.7 7.497 1.941 45 S33 ———— 0.005 ———— 4.3 7.534 1.953 46 S34

————— 0.005 ———— 4.3 7.549 1.963 47 S35 ———— 0.002 ———— 4.8 7.503 1.948 48 S36 ————

—— 0.004 3.8 7.602 1.994 49 S37 0.047 ———— 3.9 7.596 1.991 50 S38 — 0.0081 ———— 4.0 7.592

1.994

TABLE-US-00007 TABLE 7 MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES) STEEL Si + DENSITY Is No. TYPE Cu Zr Sn Sb Ce Nd Bi W Mo Nb Y sol. Al g/cm.sup.3 T 51 S39 ———— 0.055 ————

————— 4.6 7.506 1.937 52 S40 ———— 0.048 ———— 4.7 7.504 1.935 53 S41 ———— 0.004 ———— 4.7 7.512

1.938 54 S42 ———— 0.005 ———— 4.0 7.569 1.974 55 S43 ———— 0.003 ———— 4.2 7.551 1.965 56 S44 ————

—— 0.009 ———— 4.1 7.555 1.962 57 S45 ———— 0.009 ———— 4.3 7.549 1.963 58 S46 ———— 0.020 ————

4.0 7.590 1.956 59 S47 ———— 0.040 3.8 7.602 1.969 60 S48 1.800 ———— 3.9 7.598 1.961 61 S49 ————

0.0081 ———— 4.0 7.590 1.988 62 S50 ———— 0.080 ———— 4.2 7.549 1.956 63 S51 ———— 0.090 ————

—— 4.1 7.586 1.989 64 S52 ———— 0.040 ———— 4.5 7.560 1.975 65 S53 ———— 0.080 ———— 2.0 7.681 2.059

66 S54 ———— 7.2 7.317 1.838 67 S55 ———— 1.5 7.725 2.013 68 S56 ————

—— 5.0 7.379 1.898 69 S57 ———— 2.2 7.683 2.035 70 S58 ———— 2.3 7.663 2.042 71 S59

————— 2.3 7.698 2.050 72 S60 ———— 3.9 7.561 1.866 73 S61 ————

— 3.1 7.647 1.913 74 S62 5.300 ———— 3.1 7.647 1.911 75 S63 — 0.153 ———— 1.5 7.750 2.072

TABLE-US-00008 TABLE 8 MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES) STEEL Si + DENSITY Is No. TYPE Cu Zr Sn Sb Ce Nd Bi W Mo Nb Y sol. Al g/cm.sup.3 T 76 S64 ———— 0.210 ————

————— 6.0 7.384 1.893 77 S65 ———— 0.210 0.150 ———— 6.0 7.384 1.890 78 S66 ———— 0.159 — 3.3 7.558

1.990 79 S67 ———— 0.130 0.142 ———— 0.151 3.3 7.558 1.991 80 S68 ———— 0.152 0.154 ———— 3.3 7.558 1.993 81 S69 ————

————— 3.6 7.577 1.985 82 S69 ———— 3.6 7.577 1.985 83 S69 ————

3.6 7.577 1.985 84 S70 ———— 4.4 7.525 1.955 85 S70 ———— 4.4 7.525 1.955 86 S71 ————

————— 5.1 7.428 1.915

TABLE-US-00009 TABLE 9 MANUFACTURING CONDITIONS HOT BAND INTERMEDIATE ANNEALING SHEET ANNEALING FIRST COLD AVERAGE THICK- RE- ROLLING HEATING SECOND COLD NESS TEN- INTER- RATE ROLLING FINAL OF HOT TION RE-ROLL- MEDIATE RETEN- FROM ROLL- FINAL AN- ROLLED TEM- TEN- ING SHEET TION RETEN- 500° C. ING SHEET NEALING STEEL PERA- TION REDUC- THICK- TEMPER- TION TO REDUC- THICK- TEMPER- STEEL SHEET TURE TIME TION NESS ATURE TIME 650° C. TION NESS ATURE No. TYPE mm ° C. hour % mm ° C. hour ° C./sec % mm ° C. 1 S1 2.20 900 10 31.8 1.50 800 10 0.011 80.0 0.30 1050 2 S1 2.30 ———— 65.2 0.80 800 10 0.011 75.0 0.20 1180 3 S1 2.30 ———— 73.9 0.60 800 10 0.011 66.8 0.20 1080 4 S1 2.20 ———— 68.2 0.70 750 30 0.011 64.3 0.25 950 5 S1 2.20 ———— 60.9 0.86 900 10 0.011 65.1 0.30 1050 6 S1 2.30 ———— 56.5 1.00 800 10 0.011 59.9 0.40 1000 7 S1 2.19 ———— 56.5 0.95 800 10 0.011 47.4 0.50 950 8 S1 2.16 ———— 65.2 0.75 800 10 0.011 46.7 0.40 1140 9 S2 2.00 ———— 55.0 0.90 750 10 0.011 72.2 0.25 1050 10 S3 2.00 ———— 57.0 0.86 800 10 0.011 65.1 0.30 1100 11 S4 2.00 ———— 55.0 0.90 800 10 0.011 66.7 0.30 1140 12 S5 2.00 ———— 20.0 1.60 800 4 0.011 81.3 0.30 1100 13 S6 2.00 ———— 20.0 1.60 800 4 0.011 81.3 0.30 1100 14 S7 2.00 ———— 20.0 1.60 800 4 0.011 81.3 0.30 1100 15 S8 2.00 ———— 20.0 1.60 800 4 0.011 81.3 0.30 1100 16 S2 2.00 ———— 55.0 0.90 800 10 0.011 72.2 0.25 1100 17 S2 2.00 800 10 55.0 0.90 800 10 0.011 72.2 0.25 1100 18 S2 2.00 950 0.0056 55.0 0.90 800 10 0.011 72.2 0.25 1100 19 S2 2.30 1000 0.0167 70.0 0.69 1000 0.017 400 64.0 0.25 1000 20 S9 2.00 950 0.0056 55.0 0.90 800 10 0.011 72.2 0.25 1100 21 S10 2.30 1000 0.0167 70.0 0.69 1000 0.017 350 64.0 0.25 1000 22 S11 2.30 1000 0.0167 73.0 0.62 1000 0.017 350 60.0 0.25 1000 23 S9 2.30 1000 0.0167 70.0 0.69 1000 0.017 400 64.0 0.25 1000 24 S12 2.30 1000 0.0167 70.0 0.69 1000 0.017 500 64.0 0.25 1000 25 S13 2.00 1000 0.0167 70.0 0.60 1000 0.017 600 59.0 0.25 1000

TABLE-US-00010 TABLE 10 MANUFACTURING CONDITIONS HOT BAND INTERMEDIATE ANNEALING SHEET ANNEALING FIRST COLD AVERAGE THICK- RE- ROLLING HEATING SECOND COLD NESS TEN- INTER- RATE ROLLING FINAL OF HOT TION RE-ROLL- MEDIATE RETEN- FROM ROLL- FINAL AN- ROLLED TEM- TEN- ING SHEET TION RETEN- 500° C. ING SHEET NEALING STEEL PERA- TION REDUC- THICK- TEMPER- TION TO REDUC- THICK- TEMPER- STEEL SHEET TURE TIME TION NESS ATURE TIME 650° C. TION NESS ATURE No. TYPE mm ° C. hour % mm ° C. hour ° C./sec % mm ° C. 26 S14 2.00 1000 0.0167 60.0 0.80 1000 0.017 700 69.0 0.25 1000 27 S15 2.00 980 0.0167 70.0 0.60 1000 0.017 800 59.0 0.25 1000 28 S16 2.00 1000 0.0167 70.0 0.60 1000 0.017 900 59.0 0.25 1000 29 S17 2.00 1000 0.0167 70.0 0.60 1000 0.017 400 59.0 0.25 1000 30 S18 2.00 1000 0.0167 70.0 0.60 1000 0.017 500 59.0 0.25 1000 31 S19 2.00 1050 0.0167 70.0 0.60 1000 0.017 600 59.0 0.25 1000 32 S20 2.00 1050 0.0333 30.0 1.40 1050 0.008 350 82.0 0.25 980 33 S21 2.00 1000 0.0333 30.0 1.40 1050 0.008 350 82.0 0.25 980 34 S22 2.

2. The non oriented electrical steel sheet according to claim 1, wherein the chemical composition contains, by mass %, more than 3.25% and 5.0% or less of Si.
 3. The non oriented electrical steel sheet according to claim 1, wherein the chemical composition contains, by mass %, at least one of 0.0010% or more and 0.005% or less of C, 0.10% or more and less than 2.5% of sol. Al, 0.0010% or more and 3.0% or less of Mn, 0.0010% or more and 0.3% or less of P, 0.0001% or more and 0.01% or less of S, 0.0015% or more and 0.01% or less of N, 0.0001% or more and 0.10% or less of B, 0.0001% or more and 0.10% or less of O, 0.0001% or more and 0.10% or less of Mg, 0.0003% or more and 0.01% or less of Ca, 0.0001% or more and 0.10% or less of Ti, 0.0001% or more and 0.10% or less of V, 0.0010% or more and 5.0% or less of Cr, 0.0010% or more and 5.0% or less of Ni, 0.0010% or more and 5.0% or less of Cu, 0.0002% or more and 0.10% or less of Zr, 0.0010% or more and 0.10% or less of Sn, 0.0010% or more and 0.10% or less of Sb, 0.001% or more and 0.10% or less of Ce, 0.002% or more and 0.10% or less of Nd, 0.002% or more and 0.10% or less of Bi, 0.002% or more and 0.10% or less of W, 0.002% or more and 0.10% or less of Mo, 0.0001% or more and 0.10% or less of Nb, and 0.002% or more and 0.10% or less of Y.
 4. The non oriented electrical steel sheet according to claim 1, wherein the chemical composition contains, by mass %, more than 4.0% in total of Si and sol. Al.
 5. The non oriented electrical steel sheet according to claim 1, wherein the X value is 0.800 or more and less than 0.845.
 6. The non oriented electrical steel sheet according to claim 1, wherein the X value is 0.800 or more and less than 0.830.
 7. An iron core comprising the non oriented electrical steel sheet according to claim 1.
 8. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim 1.
 9. A motor comprising the iron core according to claim 7.
 10. A manufacturing method of a motor comprising a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim 1 and a process of assembling the motor using the iron core.
 11. An iron core comprising the non oriented electrical steel sheet according to claim 2.
 12. An iron core comprising the non oriented electrical steel sheet according to claim 3.
 13. An iron core comprising the non oriented electrical steel sheet according to claim 4.
 14. An iron core comprising the non oriented electrical steel sheet according to claim 5.
 15. An iron core comprising the non oriented electrical steel sheet according to claim 6.
 16. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim 2.
 17. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim 3.
 18. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim 4.
 19. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim 5.
 20. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim 6.
 21. A manufacturing method of a motor comprising a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim 2 and a process of assembling the motor using the iron core.
 22. A manufacturing method of a motor comprising a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim 3 and a process of assembling the motor using the iron core.
 23. A manufacturing method of a motor comprising a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim 4 and a process of assembling the motor using the iron core.
 24. A manufacturing method of a motor comprising a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim 5 and a process of assembling the motor using the iron core.
 25. A manufacturing method of a motor comprising a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim 6 and a process of assembling the motor using the iron core.
 26. A non oriented electrical steel sheet comprising a chemical composition containing, by mass %, 0.005% or less of C, 1.0% or more and 5.0% or less of Si, less than 2.5% of sol. Al, 3.0% or less of Mn, 0.3% or less of P, 0.01% or less of S, 0.01% or less of N, 0.10% or less of B, 0.10% or less of O, 0.10% or less of Mg, 0.01% or less of Ca, 0.10% or less of Ti, 0.10% or less of V, 5.0% or less of Cr, 5.0% or less of Ni, 5.0% or less of Cu, 0.10% or less of Zr, 0.10% or less of Sn, 0.10% or less of Sb, 0.10% or less of Ce, 0.10% or less of Nd, 0.10% or less of Bi, 0.10% or less of W, 0.10% or less of Mo, 0.10% or less of Nb, 0.10% or less of Y, and a balance comprising Fe and impurities, wherein a sheet thickness is 0.10 mm or more and 0.35 mm or less, an average grain size is 30 μm or more and 200 μm or less, an X value defined by a following expression 1 is 0.800 or more, and an iron loss $W_{\text{sub.10/1k}}$ when excited so as to be a magnetic flux density of 1.0 T at a frequency of 1 kHz is 80 W/kg or less, where the expression 1 is $X = (2 \times B_{\text{sub.50L}} + B_{\text{sub.50C}}) / (3 \times I_{\text{sub.S}})$ and where $B_{\text{sub.50L}}$ denotes a magnetic flux density in a rolling direction when magnetized with a magnetizing force of 5000 A/m, $B_{\text{sub.50C}}$ denotes a magnetic flux density in a transverse direction when magnetized with a magnetizing force of 5000 A/m, and $I_{\text{sub.S}}$ denotes a spontaneous magnetization at room temperature.
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